

HIE Knowledge Retrieval Engine: Scalable Semantic ETL & RAG Platform

B.Tech CSE (AI) Final Year | Credits: 14

1. TITLE

HIE Knowledge Retrieval Engine: A Semantic ETL and RAG-Driven Clinical Intelligence Platform

2. TEAM MEMBERS

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3. OBJECTIVE

To develop a scalable healthcare data platform that solves the problem of fragmented Electronic Health Records (EHR). The system aims to automate the ingestion, normalization, and enrichment of clinical data using a Semantic ETL backend, while providing a secure RAG layer for real-time, natural-language clinical insights.

4. LITERATURE REVIEW

Current HIEs struggle with heterogeneous data formats (HL7, CCD, XML) and manual chart review processes, which often take clinicians 15–20 minutes per patient. While traditional ETL processes handle data movement, they often lack the "semantic" layer needed for deep clinical reasoning. Recent advancements in RAG and local LLMs allow for secure, HIPAA-compliant processing of sensitive data without external API exposure, addressing the primary privacy concerns in digital health.

5. METHODOLOGY

The system integrates **Data Ingestion** via API polling, SFTP, and SQL for formats like HL7 and XML, managed by an **Orchestration** layer using APScheduler and file-locking for secure concurrency. The **Transformation** phase handles de-identification, hash-based deduplication, and memory-efficient chunking. In the **Intelligence Layer**, CCD data is parsed to JSON and mapped to LOINC standards. Finally, the **RAG Pipeline** vectorizes data into **ChromaDB**, leveraging local **Ollama (Llama 3.2)** inference to provide secure, real-time clinical summaries via a **Gradio** interface.

6. TECHNOLOGIES

The stack leverages **Python 3.6.8** and **SQL** for core logic, with **Pandas**, **APScheduler**, and **Paramiko/Boto3** driving the ETL and SFTP orchestration. Clinical intelligence is powered by **ChromaDB** and **Ollama (Llama 3.2)**, while interoperability is maintained through **Mirth Connect**, **Salesforce Health Cloud**, and **Tricare/CrossTX APIs**. The infrastructure is containerized via **Docker** on **Linux**, utilizing **AWS Secrets Manager** and **S3/Azure Blob** for secure storage, with a **Gradio** frontend for the AI interface.

7. OUTCOME

The **HIE Knowledge Retrieval Engine** delivers a high-throughput automated pipeline for clinical data via HL7, SMTP, and SMS. By streamlining the data lifecycle, the system achieves an **80% reduction in manual handling**, allowing for near-instant clinician insight retrieval—cutting search times from **20 minutes to mere seconds**.

8. SOCIAL & CLINICAL BENEFITS

The **HIE Knowledge Retrieval Engine** streamlines clinical workflows by bridging fragmented EHRs (Athena/Elation) through seamless **interoperability**. By automating chart reviews, it reduces **provider burnout** and enhances **accuracy** via LOINC/SNOMED standardization. This modular Python framework replaces costly legacy systems, delivering significant **cost efficiency** and error reduction.

9. REFERENCES

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Touvron, H., et al. (2024). Llama 3: Open Foundation and Fine-Tuned Chat Models.
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